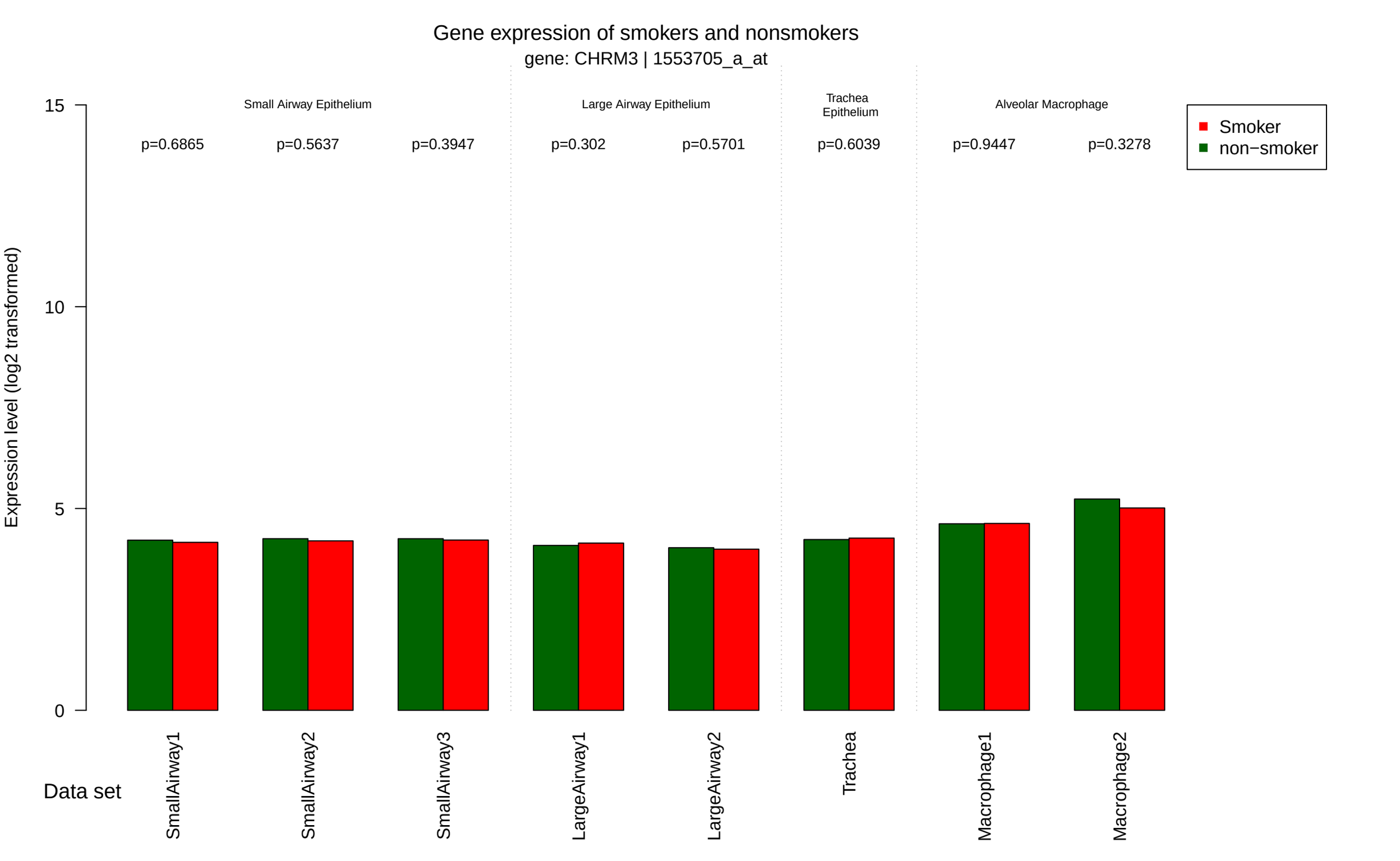
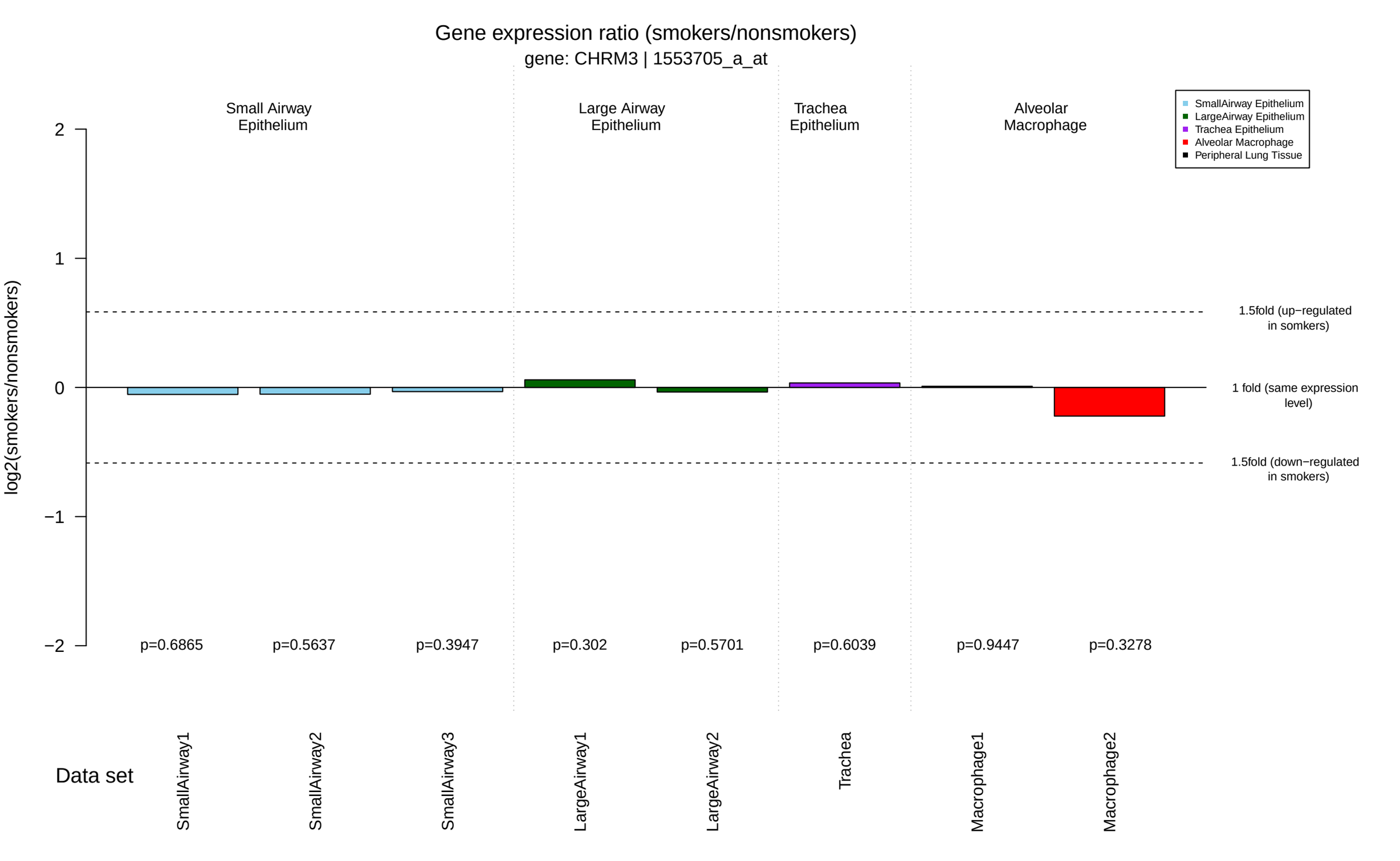
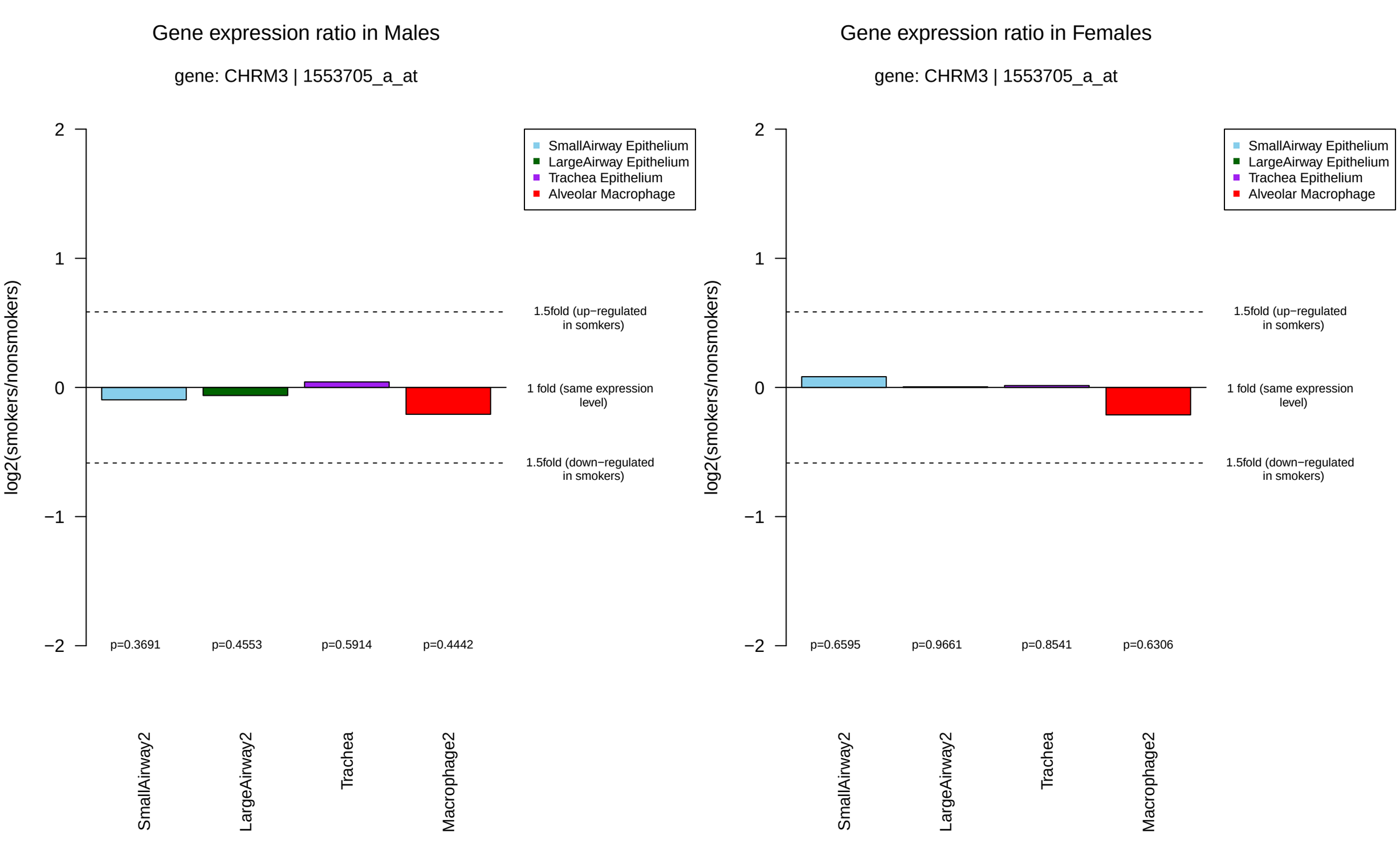
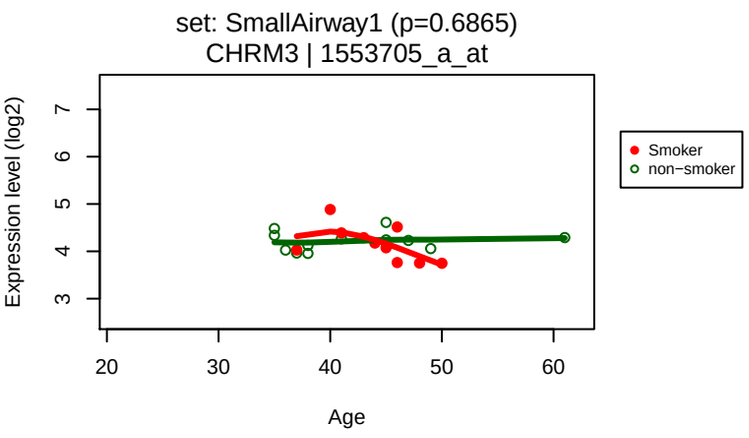
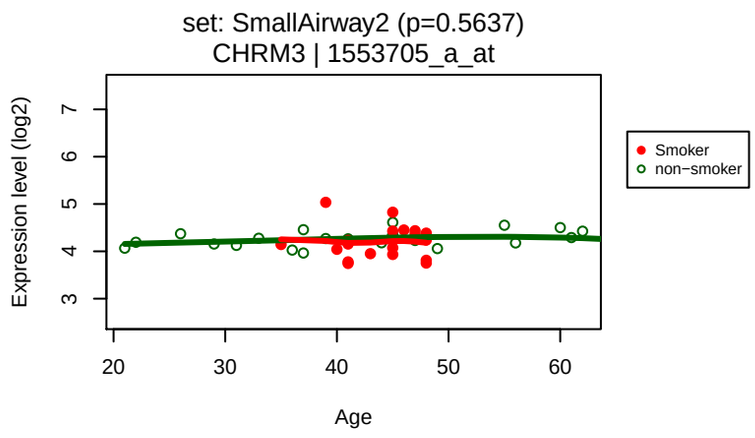
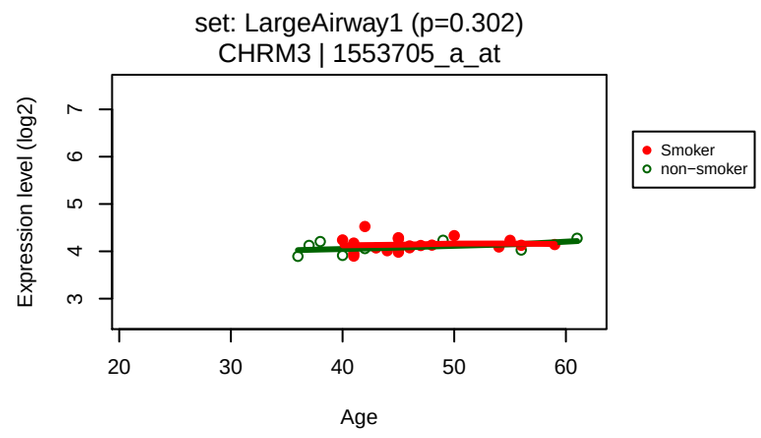

Gene expression of smokers and nonsmokers
gene: CHRM3 | 1553705_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6865
p=0.5637
p=0.3947
p=0.302
p=0.5701
p=0.6039
p=0.9447
p=0.3278
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

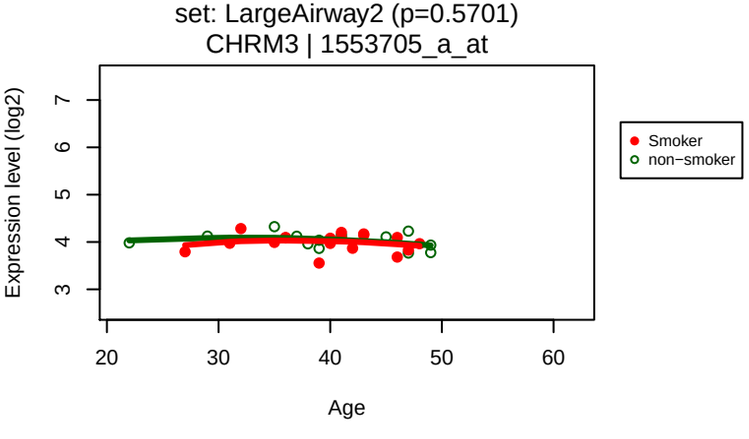

Gene expression ratio (smokers/nonsmokers)
gene: CHRM3 | 1553705_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6865
p=0.5637
p=0.3947
p=0.302
p=0.5701
p=0.6039
p=0.9447
p=0.3278
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

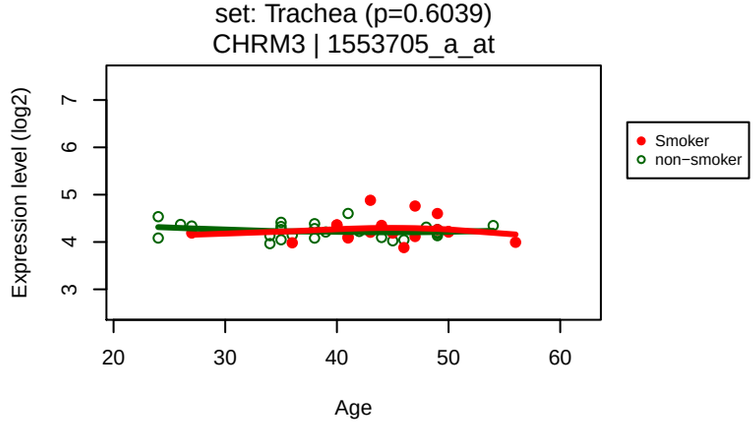

Gene expression ratio in Males
gene: CHRM3 | 1553705_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3691
p=0.4553
p=0.5914
p=0.4442
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRM3 | 1553705_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6595
p=0.9661
p=0.8541
p=0.6306
SmallAirway2
LargeAirway2
Trachea
Macrophage2

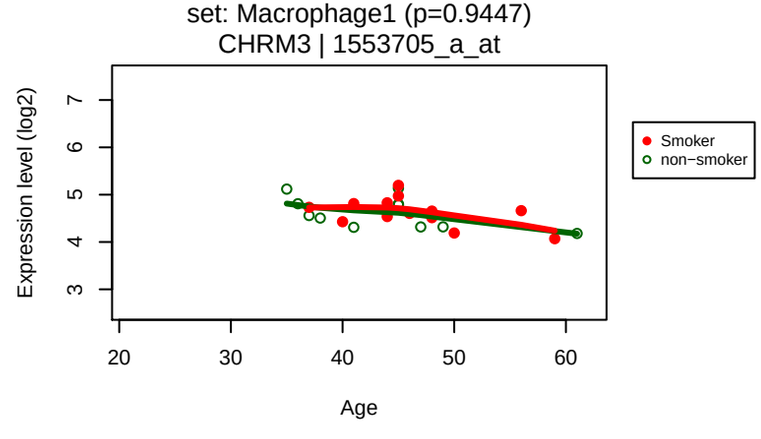

set: SmallAirway1 (p=0.6865)
CHRM3 | 1553705_a_at
Expression level (log2)
Age
Smoker
non−smoker

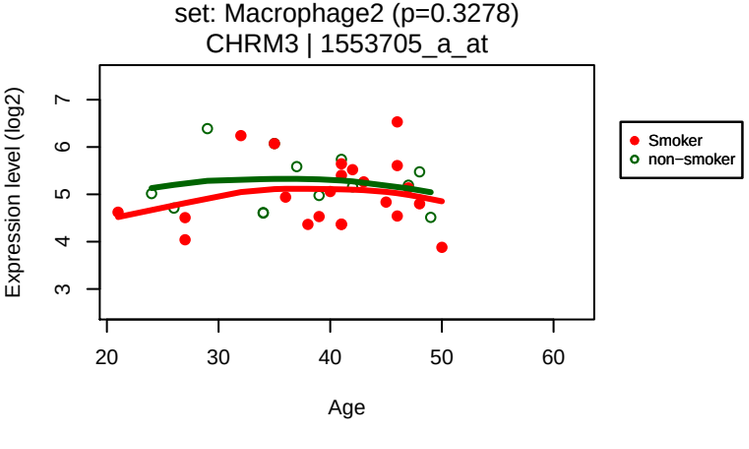

set: SmallAirway2 (p=0.5637)
CHRM3 | 1553705_a_at
Expression level (log2)
Age
Smoker
non−smoker

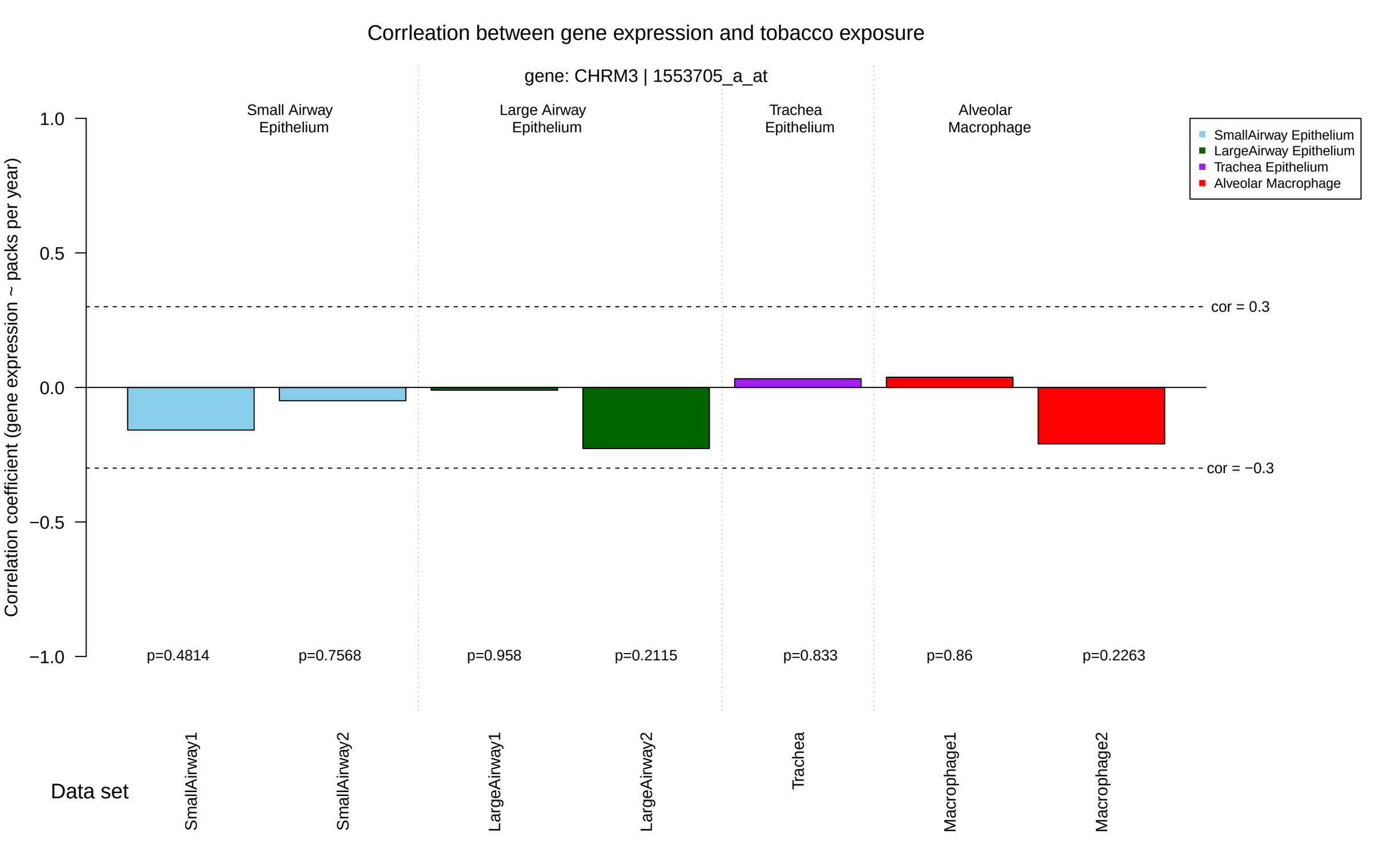

set: LargeAirway1 (p=0.302)
CHRM3 | 1553705_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5701)
CHRM3 | 1553705_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6039)
CHRM3 | 1553705_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9447)
CHRM3 | 1553705_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3278)
CHRM3 | 1553705_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRM3 | 1553705_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4814
p=0.7568
p=0.958
p=0.2115
p=0.833
p=0.86
p=0.2263
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2